

HEALTHCARE DATA CLIENT

Business Requirements Document

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Processing Scale	Approximately 250,000 records per batch
Solution Scope	Automated validation, exception reporting and Power BI visibility
Confirmed Outcome	2% exception rate; review turnaround reduced from 6 hours to 2.5 hours per file
Document status: Portfolio BRD reconstructed from confirmed engagement facts. It is not represented as the client's original signed BRD.	

1. Purpose

This BRD defines the business requirements for a healthcare data-quality solution that validates high-volume files, identifies records requiring review, produces traceable exception outputs and supports reporting visibility.

2. Business Need

Healthcare data files containing approximately 250,000 records per batch required extensive review. The business needed a repeatable way to apply data-quality checks, isolate exceptions and reduce review turnaround while keeping failed records traceable for investigation.

3. Scope

In Scope	Out of Scope / Not Claimed
Healthcare data from Excel, CSV and SQL/database sources	Exact historical database schema or table names
Claims, patients, appointments, providers, billing, transactions, clinical records and insurance data	Clinical decision-making or claim adjudication logic not confirmed by the engagement evidence
Automated data-quality checks	Unretained healthcare-specific validation thresholds or code sets
Exception flagging, reason codes and exception output	Automatic correction of every failed record
Power BI reporting visibility	Unconfirmed compliance controls or regulatory certification
Testing and documentation	Formal historical UAT sign-off records not retained

4. Business Requirements

ID	Business Requirement
BR-01	The solution shall process healthcare data files containing approximately 250,000 records per batch.
BR-02	The solution shall support data received through Excel, CSV and SQL/database sources.
BR-03	The solution shall apply repeatable data-quality checks to identify duplicate records.
BR-04	The solution shall identify null or missing values relevant to validation.
BR-05	The solution shall identify invalid dates and incorrect data formats.
BR-06	The solution shall identify invalid identifiers where validation rules are defined.
BR-07	The solution shall identify values outside defined valid ranges where applicable.
BR-08	The solution shall identify inconsistent codes where code consistency rules are defined.
BR-09	The solution shall identify cross-field mismatches based on defined relationships between fields.
BR-10	A record failing one or more validation rules shall be flagged as an exception.
BR-11	Flagged records shall be separated into an exception file, table or report for targeted review.
BR-12	Each exception shall provide a reason code or equivalent explanation of the validation failure.
BR-13	Exception records shall be visually distinguishable in the review output where highlighting is used.
BR-14	The solution shall calculate and report the exception rate for the processed batch.
BR-15	The solution shall provide reporting visibility through Power BI.
BR-16	The validation and reporting process shall support repeatable execution across incoming files or batches.
BR-17	The solution shall preserve sufficient traceability between source records, validation results and exception outputs.
BR-18	The solution shall support validation and testing of data-quality logic before reporting results.

5. Business Rules

Rule ID	Business Rule	Evidence Status
RULE-01	Exception Rate = Records failing one or more validation rules / Total records processed x 100	Confirmed definition
RULE-02	A record failing at least one applicable validation rule is treated as an exception.	Confirmed from project description
RULE-03	Exception records are routed to a separate exception file, table or report.	Confirmed
RULE-04	Exception output includes a reason code or equivalent failure explanation.	Confirmed
RULE-05	The exact field-level thresholds, valid code lists and cross-field conditions must be documented before implementation.	Portfolio governance requirement because exact historical rules are not retained

Observed result: The engagement surfaced a 2% exception rate. This is an analytical result, not a target, SLA or hard-coded threshold.

6. Functional Requirements

ID	Functional Requirement
FR-01	Ingest supported healthcare data from file or database sources.
FR-02	Execute the applicable validation checks for each batch.
FR-03	Capture validation failures at record level.
FR-04	Assign a reason code or failure description to flagged records.
FR-05	Generate an exception output for targeted analyst review.
FR-06	Calculate batch-level exception rate.
FR-07	Prepare validated data and exception results for Power BI reporting.
FR-08	Allow reviewers to distinguish valid records from records requiring attention.
FR-09	Retain traceability needed to reconcile exception output to processed source records.

7. Data Quality Requirements

Quality Dimension	Required Validation
Uniqueness	Duplicate detection
Completeness	Null and missing-value checks
Validity	Date, format, identifier and range validation
Consistency	Code consistency checks
Integrity	Cross-field mismatch checks
Traceability	Source-to-exception reconciliation and reason identification

8. Reporting Requirements

ID	Reporting Requirement
RR-01	Display processed-volume context for the batch where supported by the data.
RR-02	Display exception-rate visibility using the governed calculation.
RR-03	Provide visibility into exception categories or reason codes where source detail supports it.
RR-04	Support review of exception records without exposing fabricated or unsupported healthcare details.
RR-05	Keep reporting measures consistent with the validation output.
RR-06	Use Power BI for data-quality and exception reporting visibility.

9. Process Requirement

Step	Required Process Behaviour
1. Ingest	Receive the healthcare data from a supported source.
2. Validate	Apply defined data-quality rules.
3. Flag	Identify records failing one or more checks.
4. Classify	Attach the relevant reason code or failure explanation.
5. Output	Generate the exception file, table or report.
6. Review	Support targeted Healthcare Analyst review.
7. Report	Surface data-quality results through Power BI where required.

10. Non-Functional Requirements

ID	Requirement
NFR-01	The process should be repeatable for recurring healthcare data batches.
NFR-02	Validation logic should be documented and understandable to reviewers.
NFR-03	Exception outputs should be traceable to the processed source data.
NFR-04	The solution should handle the confirmed approximate batch scale without requiring complete manual record-by-record validation.
NFR-05	Public portfolio documentation must not expose patient identifiers, PHI or confidential client information.

11. Acceptance Criteria

AC ID	Acceptance Condition
AC-01	Supported input can be processed through the validation workflow.
AC-02	Each implemented rule produces an identifiable validation result.
AC-03	Records failing at least one rule are included in the exception output.
AC-04	Flagged records contain a reason code or equivalent failure explanation.
AC-05	Exception Rate is calculated using the governed formula.
AC-06	Exception output can be reconciled to the processed source population.
AC-07	Power BI measures reconcile with the underlying validation results.
AC-08	No fabricated patient, claim, provider or confidential client data is introduced into portfolio reporting.

12. Requirements Traceability

Business Need	BR / Rule	Downstream Evidence
Consistent high-volume validation	BR-01 to BR-09	Data Quality Rule Catalogue; Automated Validation Workflow
Targeted exception review	BR-10 to BR-13; RULE-02 to RULE-04	Exception Analysis; exception output specification
Measure data-quality outcome	BR-14; RULE-01	KPI Catalogue; Power BI Requirements
Reporting visibility	BR-15	Power BI Requirements; dashboard
Repeatability and traceability	BR-16 to BR-18	Testing; workflow; final report

13. Confirmed Business Outcome

Outcome: The solution processed approximately 250,000 records per batch, surfaced a 2% exception rate and reduced review turnaround from 6 hours to 2.5 hours per file. This equals 3.5 hours less review time per file and a derived turnaround reduction of approximately 58.3%.

Claim boundary: The 58.3% value describes review turnaround reduction only. It is not presented as a productivity increase, labor-cost saving, clinical-quality improvement or compliance outcome.